When is quantum error mitigation worth paying for?

A pre-registered study of selective error mitigation for noisy variational quantum fraud classifiers

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A noisy quantum classifier faces a spending decision



Readout mitigation

≈ 2 calibration circuits per batch



Zero-noise extrapolation

3× shots · 9× two-qubit-gate exposure



Extra shots

3× shots · removes variance, not bias

- Fraud scoring needs calibrated probabilities and reliable alerts at a fixed threshold, not just good ranking.
- Mitigation can repair noisy probabilities, but every technique costs circuits, shots and gate exposure.
- **So the operational question is: per execution context, run nothing, readout mitigation, ZNE, both, or more shots?**

Research question and hypotheses

When should a noisy QML fraud model spend additional quantum resources on mitigation, and is learning that decision better than a simple observable rule?

H-control (original)	A learned controller (Q-TRACE) using only label-free probes beats fixed policies on a reliability–cost trade-off.
H-rule (registered alternative)	A simple probe rule matches the cost-aware Oracle; learning adds no meaningful value.
Characterization	Noise mainly damages calibration; mitigation repairs calibration but not AUPRC beyond a shot-matched baseline.

Literature and novelty boundary

Work	Selects	Context used	Compared against
GSC-QEMit (Szachara et al., 2026)	abstract mitigation intensity	telemetry incl. ideal-state fidelity, simulator drift	unmitigated, static-severe
CMAB-ZNE (Rahman & Nguyen, 2026)	ZNE scale-factor sets	loss / accuracy (label-dependent)	fixed and grid-search ZNE, other QEM
APGEM (Majid et al., 2026; QRL routing)	none / ZNE / PEC / CDR / readout	fidelity, reward, noise level	fixed techniques, Oracle
Rao et al., 2026	none / DD / ZNE (fuzzy rules)	label-free circuit + hardware features	not stated in abstract
This work	**none / readout / ZNE / both / extra shots**	**label-free probes only (leakage-tested)**	**simple rules + learned, held-out, pre-registered**



Gap we address: to our knowledge, prior selectors were not tested against simple observable rules, on held-out models, under pre-registration, with extra shots and combined mitigation in the action space.

References agent-checked against publisher, arXiv and Crossref records; team confirmation pending before submission.

Dataset and leakage-safe design

Partition	Rows	Frauds	Role
VQC train	127,677	213	fit every VQC + preprocessing
VQC validation	28,372	47	qualification, thresholds, characterization
Controller train	42,559	71	fit policies (training models)
Controller validation	28,372	47	validate policies
Controller test	56,746	95	sealed; opened once



- European credit-card fraud data: 284,807 rows → 283,726 after deduplication (473 frauds, 0.17%).
- Rows identical except timestamp never straddle partitions.
- True noise parameters, labels and IDs are offline-only, enforced by a feature registry and tests.

283,726

transactions after deduplication

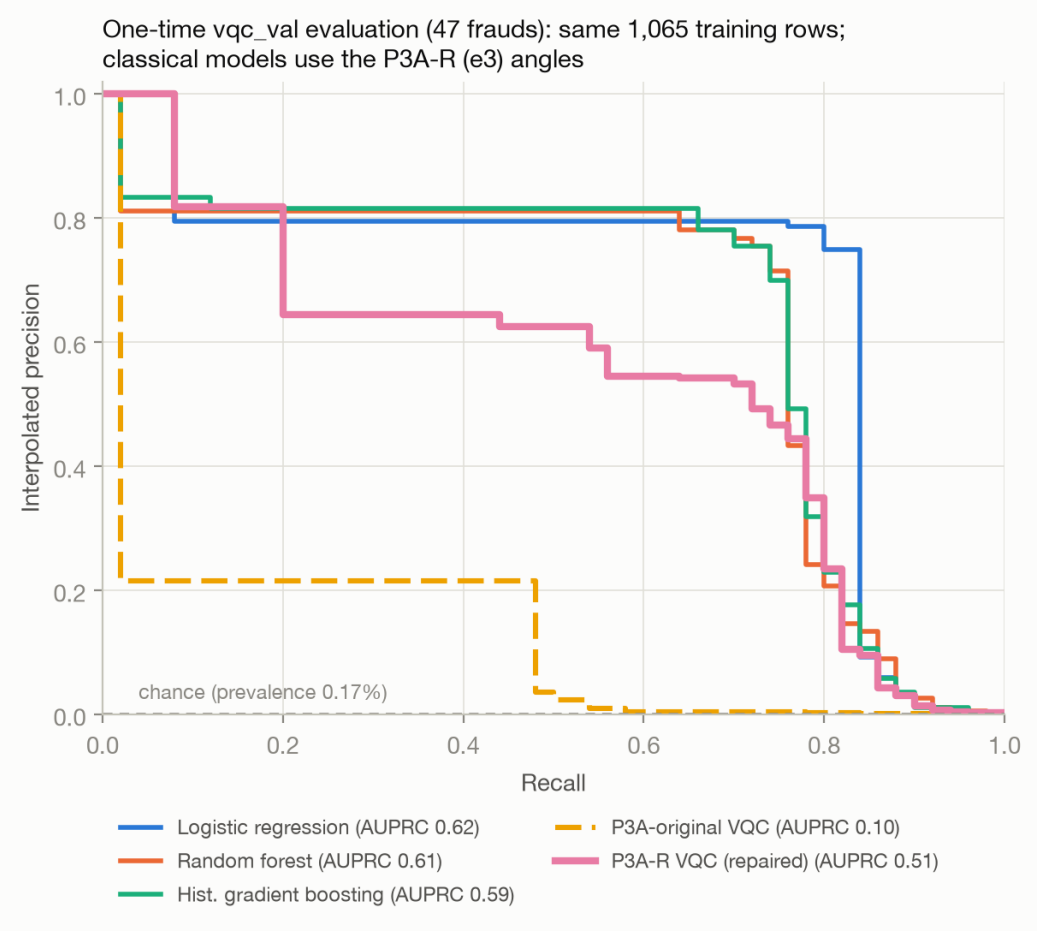
0.17%

fraud prevalence (473 frauds)

0

feature groups crossing partitions

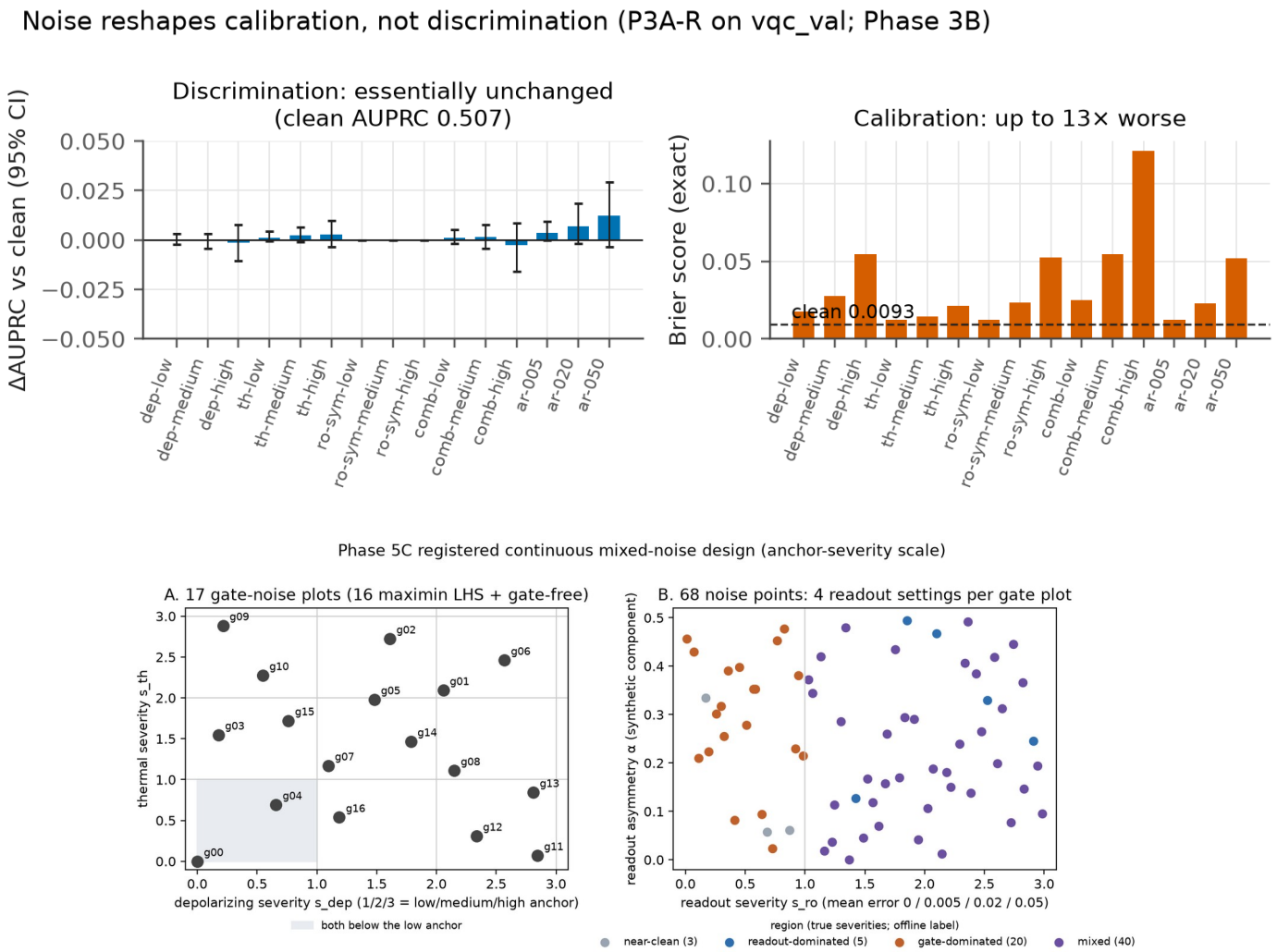
QML methodology: VQCs and a 64-model pool



Pre-registered encoding repair, validation precision–recall (Phase 3A-R).

- 4 qubits: angle encoding → ZZ feature map → RealAmplitudes / EfficientSU2; parity or mean-Z readout.
- **Encoding repair: AUPRC 0.104 → 0.507 (AUROC 0.957); classical models on the same inputs: 0.59–0.62.**
- 64 frozen VQCs = 16 structures × 4 seeds; median AUPRC 0.235 (0.047–0.538).
- Split by structure 8 / 4 / 4: test structures never seen by any policy.

Noise breaks calibration, not discrimination

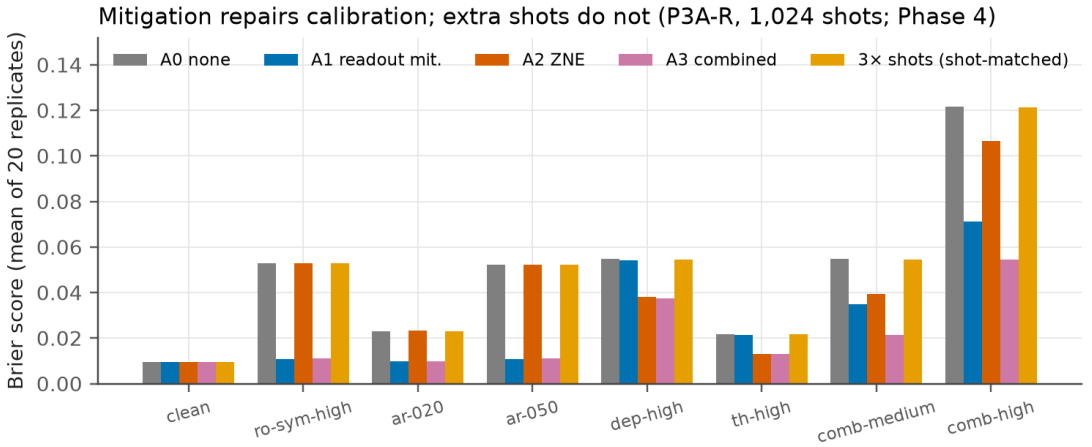


- Qiskit Aer density matrices: depolarizing, thermal relaxation, readout confusion; finite-shot draws paired across actions.
- **AUPRC moves ≤ 0.003; Brier up to 13× (0.009 → 0.121).**
- Phase 5B: 8 near-categorical conditions.
- Phase 5C: 68 continuous joint-noise points (split-plot Latin hypercube) within validated ranges; 256–4,096 shots.
- Asymmetric readout is a synthetic assumption.

Error-mitigation methods and their costs

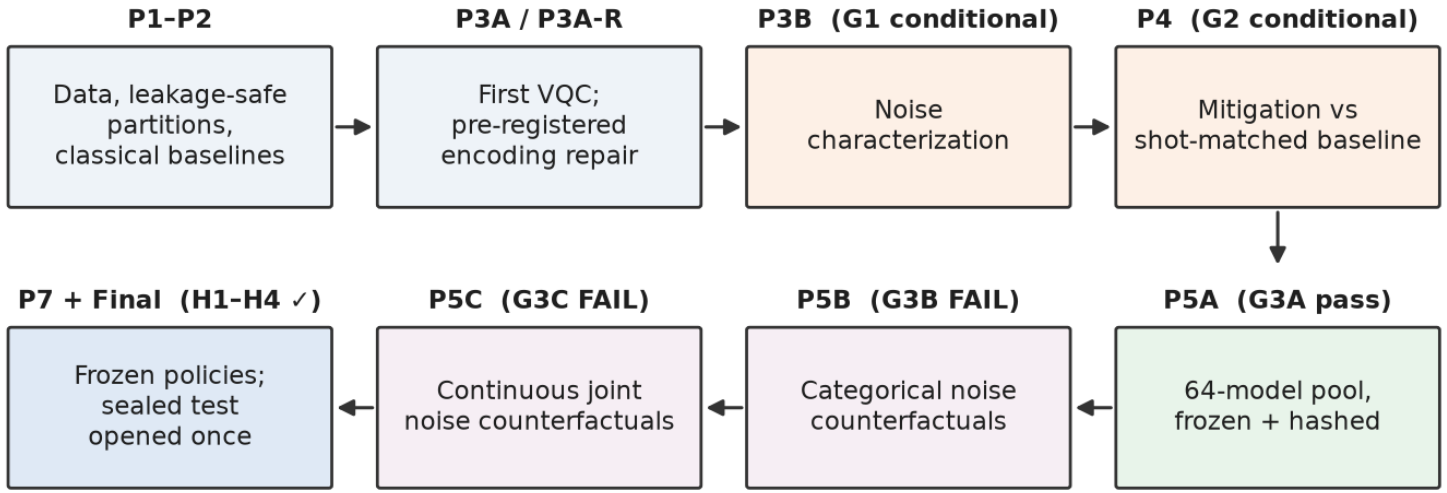
Action	Method	Shots / inference	CX exposure
A0	none	S	1×
A1	tensored readout inversion + simplex projection	S + 2S/N	1×
A4	3× shots, unmitigated	3S	3×
A2	ZNE: global folding λ = 1, 3, 5; linear fit	3S	9×
A3	readout mitigation + ZNE	3S + 2S/N	9×

S = shots per circuit; N = transactions per batch. Probe cost (66·S shots per batch) is charged to probing policies.



- No action improves AUPRC over its shot-matched baseline.
- **Brier restored: 0.053 → 0.011 (readout), 0.055 → 0.038 (ZNE), 0.122 → 0.054 (both); 3× shots: unchanged.**

Experimental design and integrity



Every phase: pre-register (config + code) → push → clean-tree reported run → validate → commit and push.

Gates decide whether to continue; controller-test outcomes stay sealed until the final protocol is pushed.

Gates decide whether to proceed; negative outcomes are reported as results.

Pre-registered

config + code pushed before every outcome

Sealed test

unlocked once, by a guard; reads logged

Frozen comparators

decision fingerprints verified at test

Reproducible

16 slow tests re-run Phases 1–5C and the final test

Controllers, baselines and two failed gates

Policies compared (all fitted on training models)

- Never / always A1–A4; best single action
- R_cat: 5B category thresholds (0.005, 0.90)
- **R_thr: 2 probe thresholds per budget (deployable)**
- R_obs: R_thr per readout type
- Learned (fixed specs): logistic, depth-3 tree, gradient boosting; GBT predicting per-action Brier
- Oracle: cheapest action within 10% of best Brier

99.7%

G3B (categorical noise): a 4-cell probe rule recovers the best action. FAIL → no controller.

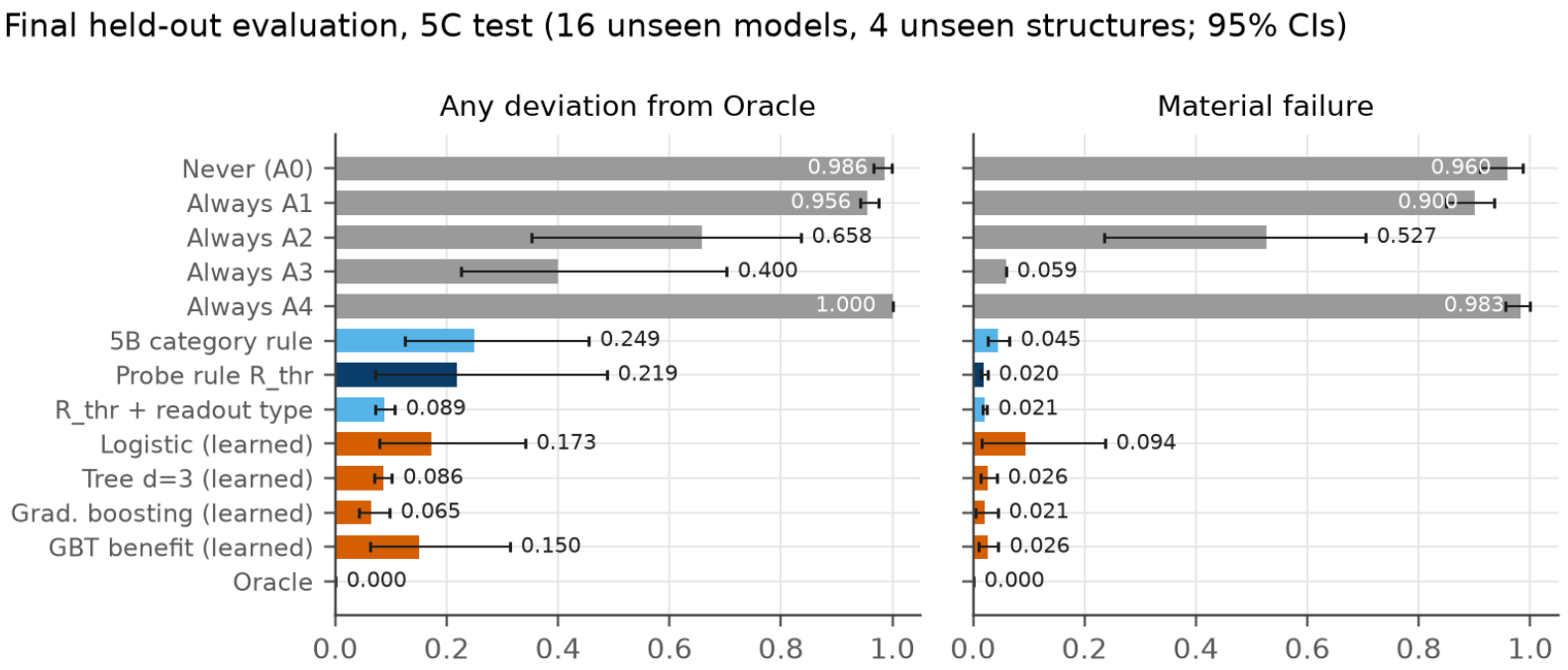
2.0%

G3C (continuous joint noise): material errors of the best rule [1.6, 2.7]. FAIL → no controller.

0%

contexts where 3× shots was the best action, in either environment.

Statistical results on the sealed held-out test



2.0%

material errors, R_thr (95% CI 1.5–2.6%), 16 unseen models

−0.03%

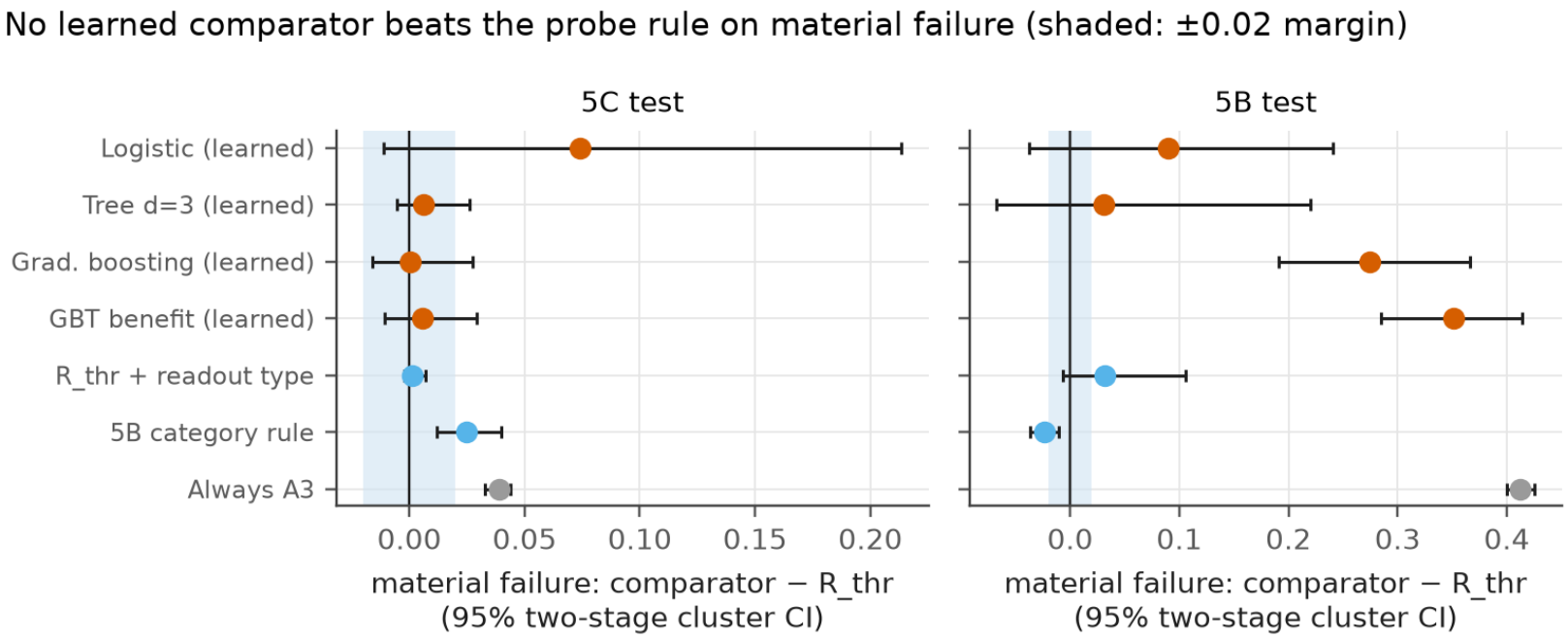
Brier regret vs Oracle (CI −1.7 to +1.2%)

8.8%

material errors under categorical noise (5B test; H4 ≤ 10%)

Two-stage cluster bootstrap over only 4 held-out structures, so CIs are likely too narrow. H1–H4 all supported.

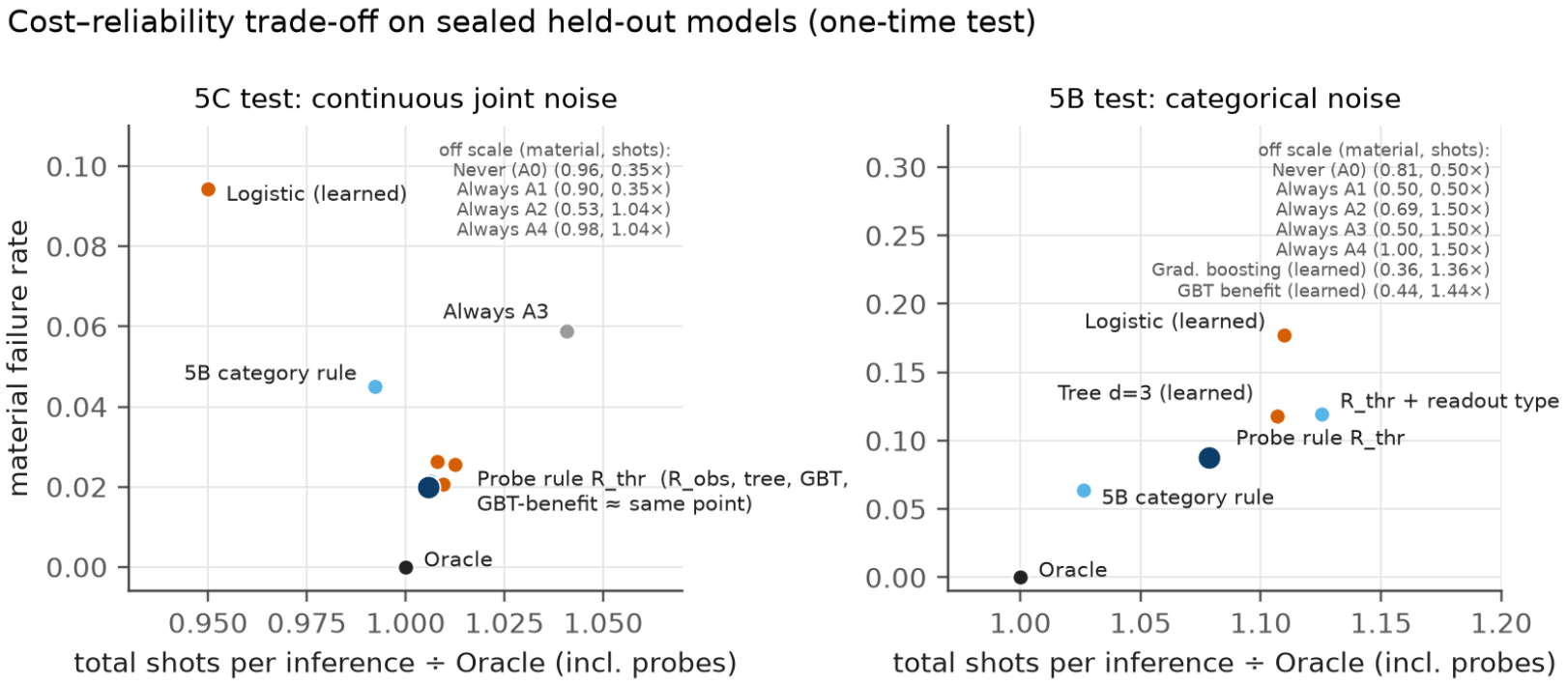
No learned comparator reduces material errors



- **Best learned on 5C test (gradient boosting): 2.1% vs 2.0%; difference +0.0005 [−0.016, +0.027].**
- It agrees with the Oracle more often (6.5% vs 21.9% disagreement); the rule's extra choices are cheap.
- Categorical noise: learned 12–44%; R_cat beats R_thr (6.4% vs 8.8%).
- Unseen mean-Z structure: learned 5–30%, rule 1.4%.

Comparator − R_thr; positive favours the rule; shaded band = pre-registered ±0.02 margin.

Cost–reliability trade-offs



+0.6%

shots vs the Oracle (R_thr, 5C test)

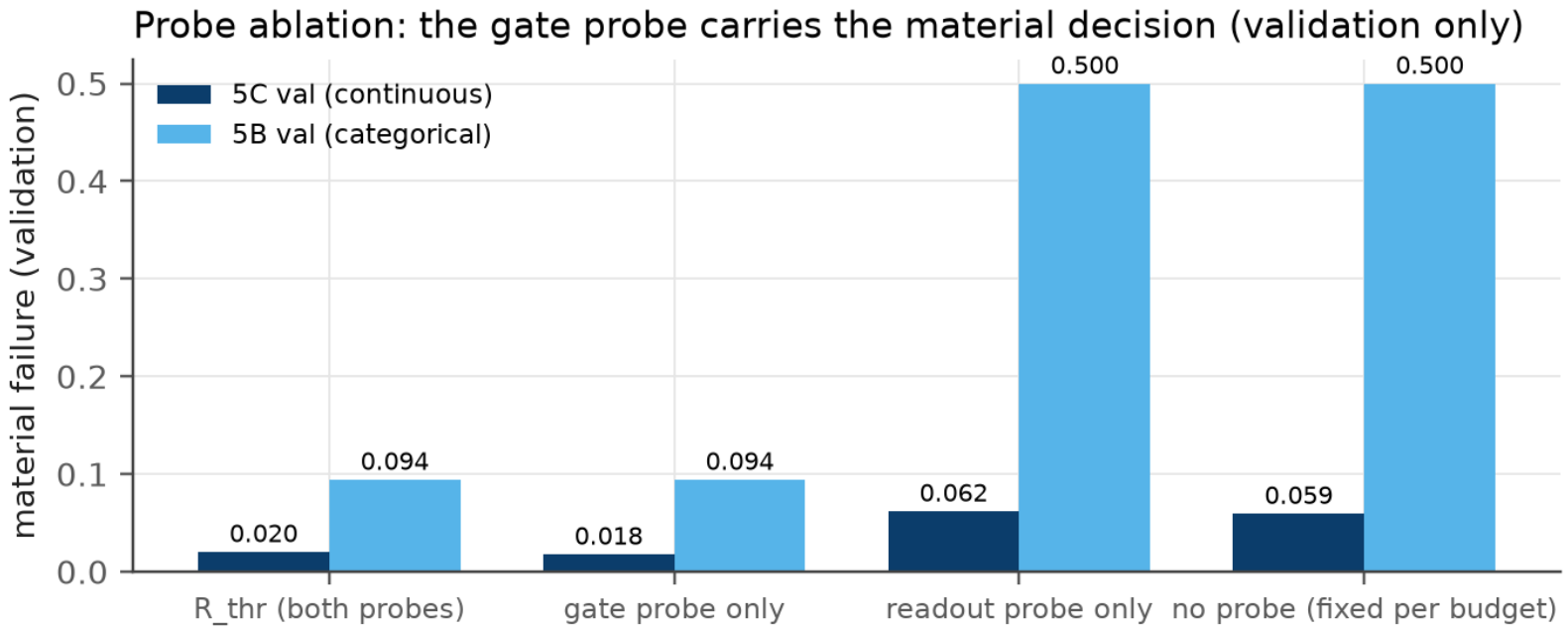
−3.4%

shots vs always-combined; its 5.9% errors = the gate-free share of our design

0.35×

shots for never-mitigating, at 96% material errors

Ablations: the gate probe carries the decision



1.8%

material errors with the gate probe alone (5C validation)

≈ 645–685

batch size below which probing does not pay (5C; depends on gate-free share)

- Shot budget changes the best action in < 3% of cases.
- Tolerance 0.05–0.20: rule material errors 2.0–7.5%.

Validation data only; the sealed test was not reused.

Limitations and threats to validity

Simulation only

No hardware run; asymmetric readout is synthetic; no crosstalk, drift or coherent errors.

Small scale

4 qubits, shallow circuits, one public dataset (0.17% fraud, inherited PCA).

No quantum advantage

Classical models score higher on the same inputs.

Narrow mitigation set

Readout inversion and linear ZNE only; no PEC, CDR or dynamical decoupling.

Probe and design

Gate probe barely sees thermal noise (Spearman −0.12); 16 of 17 gate settings carry gate noise.

Few held-out structures

Only 4: CIs likely too narrow; thresholds tuned on clean validation data.

Conclusions and future research

What we found

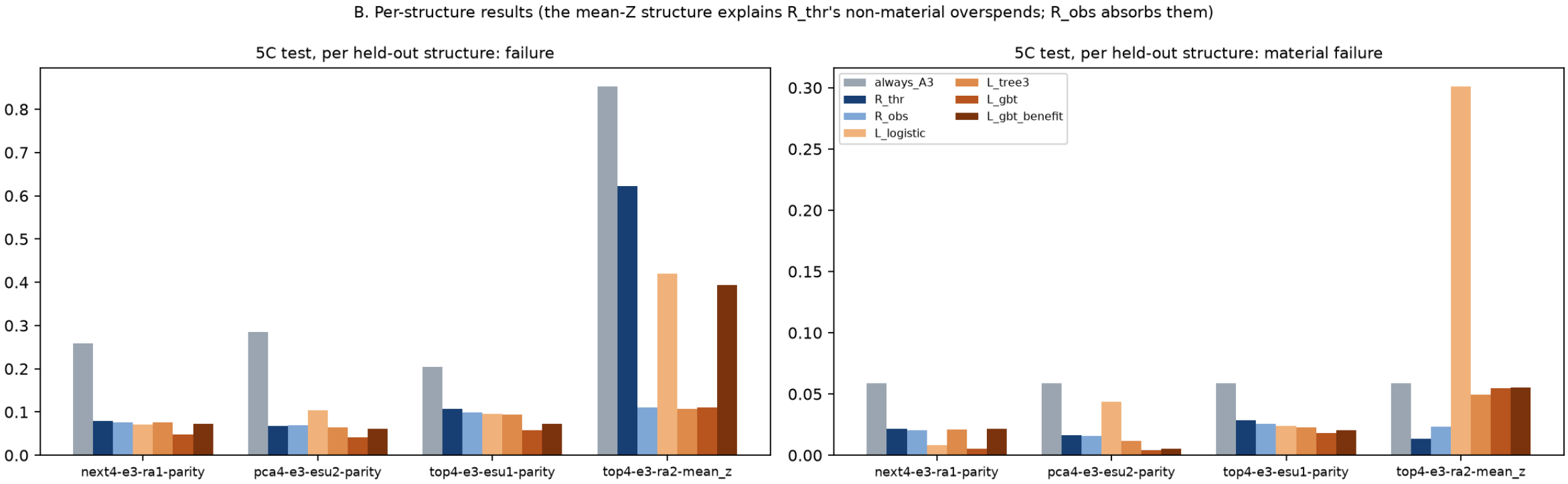
- Mitigation fixes calibration, not discrimination; extra shots never substitute.
- In our joint-noise design the material decision is: is gate noise present?
- A two-threshold probe rule is near-Oracle on unseen models; fixed learned comparators did not reduce material errors.

Next questions

- Hardware validation with real readout asymmetry and drift.
- Richer mitigation (PEC, CDR, DD) and a thermal-sensitive probe.
- Larger circuits, where noise may also degrade discrimination.

Answer: in the studied setting, a simple probe rule performed competitively with learned controllers.

Backup: per-structure held-out results



The unseen mean-Z structure explains R_thr's higher raw disagreement (non-material: two calibration circuits); R_obs absorbs it.

Backup: Phase 5C preferred-action map

